

1. 利用已知的初等函数的展开式,求下列函数在 $x=0$ 处的幂级数展开式,并指出收敛域:

- (1) $\frac{x}{16+x^2}$;
- (2) e^{x^2} ;
- (3) $\frac{1}{a+x} \ (a \neq 0)$;
- (4) $\ln \sqrt{\frac{1+x}{1-x}}$;
- (5) $(1+x)e^{-x}$;
- (6) $\frac{1}{2}(e^x - e^{-x})$;
- (7) $\sin^3 x$ (提示: $\sin^3 x = \frac{1}{4}(3\sin x - \sin 3x)$);
- (8) $\sin\left(\frac{\pi}{4} + x\right)$;
- (9) $\ln(1+x-2x^2)$;
- (10) $\frac{5x-12}{x^2+5x-6}$.

2. 利用逐项微分法和逐项积分法求下列函数在 $x=0$ 处的幂级数展开式

(2) $e^{x^2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$. $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = 0$, 收敛域为 $\mathcal{R}$

(3) $\frac{1}{a+x} = \frac{1}{a} \cdot \frac{1}{1+\frac{1}{a}x} = \frac{1}{a} \cdot \sum_{n=0}^{\infty} (-1)^n \left(\frac{x}{a}\right)^n \quad x \in (-a, a)$
 $= \sum_{n=0}^{\infty} (-1)^n \frac{x^n}{a^{n+1}}$

(4) $\frac{1}{2} \ln \frac{1+x}{1-x} = \frac{1}{2} (\ln(1+x) - \ln(1-x)) = \frac{1}{2} \left(\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n + \sum_{n=1}^{\infty} \frac{1}{n} x^n \right)$
 $= \frac{1}{2} \cdot \sum_{n=1}^{\infty} (1+(-1)^{n+1}) \cdot \frac{x^n}{n}$
 $= \sum_{n=1}^{\infty} \frac{x^{2n-1}}{2n-1}, \quad x \in (-1, 1)$

(5) $e^{-x} + xe^{-x} = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!} + x \sum_{n=0}^{\infty} \frac{(-x)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n (x^n + x^{n+1})}{n!} = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!} + \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n!}, \quad x \in \mathcal{R}$

(6). $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, $e^x - e^{-x} = \sum_{n=0}^{\infty} \frac{1}{n!} \cdot (1 - (-1)^n) \cdot x^n = 2 \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$
 $\therefore \text{奇数项} = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}, \quad x \in \mathcal{R}$

(7) $\sin^3 x = \frac{1}{4} (3\sin x - \sin 3x) = \frac{3}{4} \cdot \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-1}}{(2n-1)!} - \frac{1}{4} \sum_{n=1}^{\infty} \frac{(3x)^{2n-1} (-1)^n}{(2n-1)!}, \quad x \in \mathcal{R}$
 $= \frac{1}{4} \sum_{n=1}^{\infty} \frac{(3-3^{2n-1}) \cdot x^{2n-1}}{(2n-1)!} \cdot (-1)^n, \quad x \in \mathcal{R}$

(8) $\frac{\pi}{4} + x = t, \quad x = t - \frac{\pi}{4}, \quad \sin\left(\frac{\pi}{4} + x\right) = \sin t = \sum_{n=1}^{\infty} \frac{t^{2n-1}}{(2n-1)!}$

$\sin\left(\frac{\pi}{4} + x\right);$

令 $\frac{\pi}{4} + x = t$, 则 $x = t - \frac{\pi}{4}$, $\sin(x + \frac{\pi}{4}) = \sin t = \sum_{n=1}^{\infty} \frac{t^{2n-1}}{(2n-1)!}$
 $= \sum_{n=1}^{\infty} \frac{(x + \frac{\pi}{4})^{2n-1}}{(2n-1)!}$

(不是在0处的泰勒展开)

(8) $\sin(\frac{\pi}{4} + x) = \frac{\sqrt{2}}{2} (\sin x + \cos x) = \frac{\sqrt{2}}{2} \left(\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} + \sum_{n=0}^{\infty} \frac{x^{2n} (-1)^n}{(2n)!} \right)$
 $= \frac{\sqrt{2}}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{x^{2n+1}}{(2n+1)!} + \frac{x^{2n}}{(2n)!} \right)$
 $x \in \mathcal{R}$

- (8) $\sin\left(\frac{\pi}{4} + x\right);$
- (9) $\ln(1+x-2x^2);$
- (10) $\frac{5x-12}{x^2+5x-6}.$

(9) $\ln(1-x)(2x+1) = \ln(1-x) + \ln(2x+1)$
 $= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (-x)^n}{n} + \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (2x)^n}{n}$
 $= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (2x)^n - x^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} 2^n - 1}{n} x^n$

对 $\ln(1-x), \quad x \in [-1, 1)$

对 $\ln(2x+1), \quad x \in (-\frac{1}{2}, \frac{1}{2}]$

$\therefore x \in (-\frac{1}{2}, \frac{1}{2}]$

$$(10) \text{ 设 } \frac{5x-12}{(x+6)(x-1)} = \frac{A}{x+6} + \frac{B}{x-1} = \frac{(A+B)x-A+6B}{(x+6)(x-1)} \quad \therefore \begin{cases} A+B=5 \\ 6B-A=-12 \end{cases} \Rightarrow \begin{cases} A=6 \\ B=-1 \end{cases}$$

$$\therefore \text{原式} = \frac{6}{x+6} + \frac{1}{1-x} = \frac{1}{1+\frac{1}{6}x} + \frac{1}{1-x} = \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{6}x\right)^n + \sum_{n=0}^{\infty} x^n \\ = \sum_{n=0}^{\infty} \left[(-1)^n \cdot \frac{1}{6^n} + 1\right] x^n$$

2. 利用逐项微分法和逐项积分法, 求下列函数在 $x=0$ 处的幂级数展开式:

(1) $\arctan x$; (2) $\ln(x + \sqrt{1+x^2})$; (3) $\int_0^x \frac{dx}{\sqrt{1-x^4}}$.

$$(1) \quad f(x) = \arctan x, \quad f'(x) = \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}, \quad \therefore f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$$

$$(2) \quad f(x) = \ln(x + \sqrt{1+x^2}), \quad f'(x) = \frac{1}{\sqrt{1+x^2}} = (1+x^2)^{-\frac{1}{2}} = 1 + \left(-\frac{1}{2}\right)x^2 + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!}x^4 + \dots + \frac{(-1)^n \cdot (2n-1)!!}{n!} x^{2n} + \dots \\ = \sum_{n=1}^{\infty} \frac{(-1)^n \cdot (2n-1)!!}{(2n)!!} x^{2n} + 1$$

$$f(x) = x + \sum_{n=1}^{\infty} \frac{(-1)^n \cdot (2n-1)!!}{(2n)!! \cdot (2n+1)} x^{2n+1}$$

$$(3) \quad \int_0^x \frac{dx}{\sqrt{1-x^4}} = f(x), \quad f'(x) = \frac{1}{\sqrt{1-x^4}} = (1-x^4)^{-\frac{1}{2}} = \sum_{n=1}^{\infty} \frac{(2n-1)!!}{(2n)!!} x^{4n} + 1$$

$$f(x) = x + \sum_{n=1}^{\infty} \frac{(2n-1)!!}{(2n)!! \cdot (4n+1)} x^{4n+1}$$

3. 证明级数 $\sum_{n=0}^{\infty} \frac{x^n}{(n+1)!} = \frac{1}{x}(e^x - 1)$ ($x \neq 0$), 并证明

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!} = 1.$$

$$e^x - 1 = \sum_{n=1}^{\infty} \frac{x^n}{n!}, \quad \frac{1}{x}(e^x - 1) = \sum_{n=1}^{\infty} \frac{x^{n-1}}{n!} = \sum_{n=0}^{\infty} \frac{x^n}{(n+1)!}$$

$$\therefore \sum_{n=0}^{\infty} \frac{1}{(n+1)!} = e - 1$$

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!} = \sum_{n=1}^{\infty} \frac{n+1-1}{(n+1)!} = \sum_{n=1}^{\infty} \frac{1}{n!} - \sum_{n=1}^{\infty} \frac{1}{(n+1)!} \\ = e - 1 - e + 2 = 1$$

4. 仿照上题的办法, 证明:

$$\sum_{n=1}^{\infty} \frac{n}{(n+2)!} = 3 - e.$$

5. 求下列定积分的近似值, 准确到 10^{-3} :

(1) $\int_0^2 \frac{\sin x}{x} dx$; (2) $\int_0^1 \cos x^2 dx$.

$$\sum_{n=1}^{\infty} \frac{n+2-2}{(n+2)!} = \sum_{n=1}^{\infty} \frac{1}{(n+1)!} - 2 \sum_{n=1}^{\infty} \frac{1}{(n+2)!} = e - 2 - 2 \cdot \left(e - \frac{5}{2}\right) = 3 - e$$

$$5. \quad f(x) = \int_0^x \frac{\sin x}{x} dx, \quad f'(x) = \frac{\sin x}{x} = \frac{1}{x} \cdot \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)!} = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-2}}{(2n-1)!}$$

$$\therefore f(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)! \cdot (2n-1)}$$

$$\int_0^2 \frac{\sin x}{x} dx = f(2) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{2^{2n-1}}{(2n-1)! \cdot (2n-1)} = 2 + (-1) \cdot \frac{8}{6 \times 3} + \frac{32}{120 \times 5} \dots$$

$$|x_n| \leq 10^{-3}$$